MILITARY SPECIFICATION

TEST SET, RADIO, AN/ARM-156

This specification has been approved by the
Naval Air Systems Command, Department of the Navy

1. SCOPE

1.1 Scope - This specification covers one type of equipment designated as Test Set, Radio, AN/ARM-156 intended for intermediate and depot levels of use in support of AN/ARN-21, AN/ARN-52, AN/ARN-65, AN/ARN-84, AN/ARN-84(V), AN/ARN-86, AN/ARN-91, AN/ARN-103, AN/ARN-105, AN/ARN-113, AN/ASQ-19, AN/ASQ-56, AN/ASQ-57, AN/ASQ-58, AN/ASQ-88, TACAN Stationkeeping, TACAN Data Link, Inverse TACAN or equivalent airborne TACAN systems. See paragraph 3.5.2.4.1.2.4.

1.2 Classification - The Test Set, Radio, AN/ARM-156 shall be of one type, class and shall consist of the following major components:

Quantity Per Equipment	Item	Designation	Appl Para
1	Power Supply - Video Pulse	PP-6730/ARM-156	3.5.1
1	RF Signal Gen - Power Meter	SG-980/ARM-156	3.5.2
1	Transit Cases	*	3.5.3
1 Set	Cables and Accessories	*	3.5.4

*Official designations have not been assigned.

1.3 Associated Equipment - This equipment shall operate with the associated equipment listed in paragraph 6.8.

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids form a part of this specification to the extent specified herein:

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SPECIFICATIONS

Military

MIL-C-3432	Cable, Flexible, 300 and 600 volts
MIL-C-3767	Connectors (Electrical, Power, Bladed Type)
MIL-D-18300	Design Data for Avionic Equipment
MIL-E-17555	Electrical and Electronic Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-M-7793	Meter, Time Totalizing
MIL-N-18307	Nomenclature and Nameplates for Aeronautical Electronic and Associated Equipment
MIL-N-81207(AS)	Navigation Set, TACAN, AN/ARN-84(*)
MIL-N-22239	Navigational Set, TACAN, AN/ARN-52(V)
MIL-R-7060	Radio Set, AN/ARN-21
MIL-R-83214(USAF)	Radio Set, AN/ARN-65(V)
MIL-R-22763(WEPS)	Receiver-Transmitter, Radio, RT-547(*)/ASQ-19 and Decoder, Pulse, KY-312(*)/ASQ-19
MIL-R-22677(WEPS)	Receiver-Transmitter, Radio, RT-541(*)/ASQ and Decoder, Pulse, KY-309(*)/ASQ. (AN/ASQ-56, -57, -58)
MIL-R-81105(WEPS)	Receiver-Transmitter, Radio RT-736(*)/ASQ-88 and Decoder, Pulse, KY-531(*)/ASQ-88
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment, Format for
MIL-T-21200	Test Equipment for use with Electronic and Fire Control Systems, General Specification for

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Standards - Military

MIL-STD-704	Electric Power, Aircraft Characteristics and Utilization of
MIL-STD-287	Test Signals for Electronic Tactical Navigation Equipment (Revised to Include Y Mode Parameters)
MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electronic and Electronic Equipments Design Standard for
MIL-STD-291	Military Standard, Standard Tactical Air Navigation (TACAN) Signal
MIL-STD-415	Test Points and Test Facilities for Electronic Systems and Associated Equipment
MIL-STD-471	Maintainability Demonstration
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-463	Definitions and System of Units, Electromagnetic Interference Technology
MIL-STD-461	Electromagnetic Interference Characteristics, Requirement for
MIL-STD-781	Reliability Test Exponential Distribution
MIL-STD-470	Maintainability Program Requirements
MIL-STD-721	Definitions, Effectiveness Terms
MS-2548	Cable Assembly, Power Electrical

Naval Air Systems Command

WR-97	Aircraft Electronics Equipment Reliability Requirements for Design of
AR-5	Microelectronic Devices Used in Avionic Equipment, Procedures for Selection and Approval of
AR-10	Maintainability of Avionic Equipment and Systems, General Requirements for

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2.1.1 Availability of Documents - When requesting specifications, standards, drawings or publications, refer to both title and number. Copies of this specification and applicable specifications required by contractors in connection with specific procurement functions may be obtained upon application to the Commanding Officer, Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

3. REQUIREMENTS

3.1 Testing - This specification makes provisions for testing (see 4).

3.2 Parts, Materials and Processes - The materials, parts and processes shall be in accordance with MIL-T-21200.

3.2.1 Selection of Parts, Materials, and Processes - Except as specified herein, MIL-T-21200 shall apply for the selection of parts, materials and processes.

3.2.2 Approval of Nonstandard Parts and Materials - Approval for the use of nonstandard parts and materials shall be in accordance with MIL-T-21200.

3.3 Design and Construction - The design and construction of the equipment shall be in accordance with MIL-T-21200, Class 3.

3.3.1 Total Weight - The total weight of the equipment including cables shall not exceed 145 pounds.

3.3.2 Reliability -

3.3.2.1 Operational Stability - The equipment shall operate with satisfactory performance, continuously or intermittently for a period of at least 300 hours without the need to readjust any of the controls which are not accessible from the front panel.

The equipment shall meet the stability requirements under the following conditions:

3.3.2.1.1 Unstacked configuration (i.e., side by side with associated equipment, test set, indicator).

3.3.2.1.2 Stacked configuration (i.e., in a vertical column with associated equipment, test set, indicator).

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3.3.2.2 Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring scheduled replacement due to wear during the life of the equipment and the wearout life of such parts, shall be determined by the contractor and submitted to the procuring agency.

3.3.2.3 Reliability in MTBF (Mean Time Between Failures) - The equipment shall have 1330 hours of mean (operating) time between failures when tested and accepted as outlined under the requirements of Specification MIL-T-21200 and Military Standard 781.

3.3.2.4 Time Totalizing Meter - The equipment shall contain a time totalizing meter on the control panel in accordance with MIL-M-7793.

3.3.3 Cabling and Connections

3.3.3.1 Cabling and Connectors - The equipment shall provide for the use of cables and connectors in accordance with the requirements of Specification MIL-T-21200. Insofar as practical, all control panel connectors shall be mounted along the lower edge of the control panel and in no case shall their location be such that the connecting cables will interfere with the operator.

3.3.3.2 Cable Assemblies, Power Electrical - Applicable input power cable and connections shall conform to MIL-T-21200. Adapters shall be provided for ship and shore shop uses.

3.3.4 Interchangeability - The equipment shall meet the interchangeability requirements as defined in Specification MIL-T-21200.

3.3.5 Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Military Standards 461, 462 and 463. Also see paragraph 3.4.2.3.

3.3.6 Self-Test Capability - The test set shall contain an interruptive self-test cycle at all positions of the frequency selector switch. Visual indications of range delay, bearing delay and proper power meter operations shall be displayed during the interruptive self-test cycle.

Additionally, during test set operation, the test set shall include the capability of self-check and display the following functions:

Range
Range Rate
Bearing
Bearing Rate

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Antenna Speed
15 Hz Phase Shift
A/A Pulse Repetition Frequency
Test Set Reply Efficiency - (This test requires interrogation by airborne set under test)
% Modulation - 15 Hz
% Modulation - 135 Hz
RF Output Level

3.3.7 Maintainability - Maintainability shall be in accordance with MIL-T-21200, and AR-10 to the extent specified herein:

3.3.7.1 Modularization - The equipment shall be modularized at all levels of maintenance. The weight and Mean Time Between Failure of the equipment units shall meet the requirements of 3.3.1 and 3.3.2.3, respectively. Circuits shall be packaged into discrete replaceable modules of such cost and reliability that disposal-on-failure rather then module repair is the most cost effective logistics support action. The criteria for disposal-on-failures shall be as specified in figure 1.

3.3.7.2 Standardization - The number of different types of critical parts shall be kept to a minimum in accordance with MIL-T-21200 and AR-10.

3.3.7.3 Encapsulation and Conformal Coatings - Modules or assemblies shall meet the requirements of AR-10, paragraph 3.1.5.

3.3.7.4 BIT (Built-in Test) Function - BIT features shall be as specified in paragraph 3.3.6 and 3.4.8. The BIT capability in association with operator observation of front panel displays, test procedures, and utilization of common and standard test equipment shall provide for detection of an equipment malfunction and isolation of the malfunction to a replaceable assembly.

3.3.7.5 Test Points - Test points shall be provided for all levels of maintenance and shall provide safe and convenient access to electronic circuits. The test points provided shall enable the operator to determine the performance level of the equipment unit and in the event of a failure, to ascertain which replaceable assembly contained therein is at fault.

3.3.7.6 Shop Fixed Interface - Replaceable assemblies shall not require adjustment or trimming upon installation into an equipment unit with the exception of the amplitude and pulse modulator replaceable assemblies.

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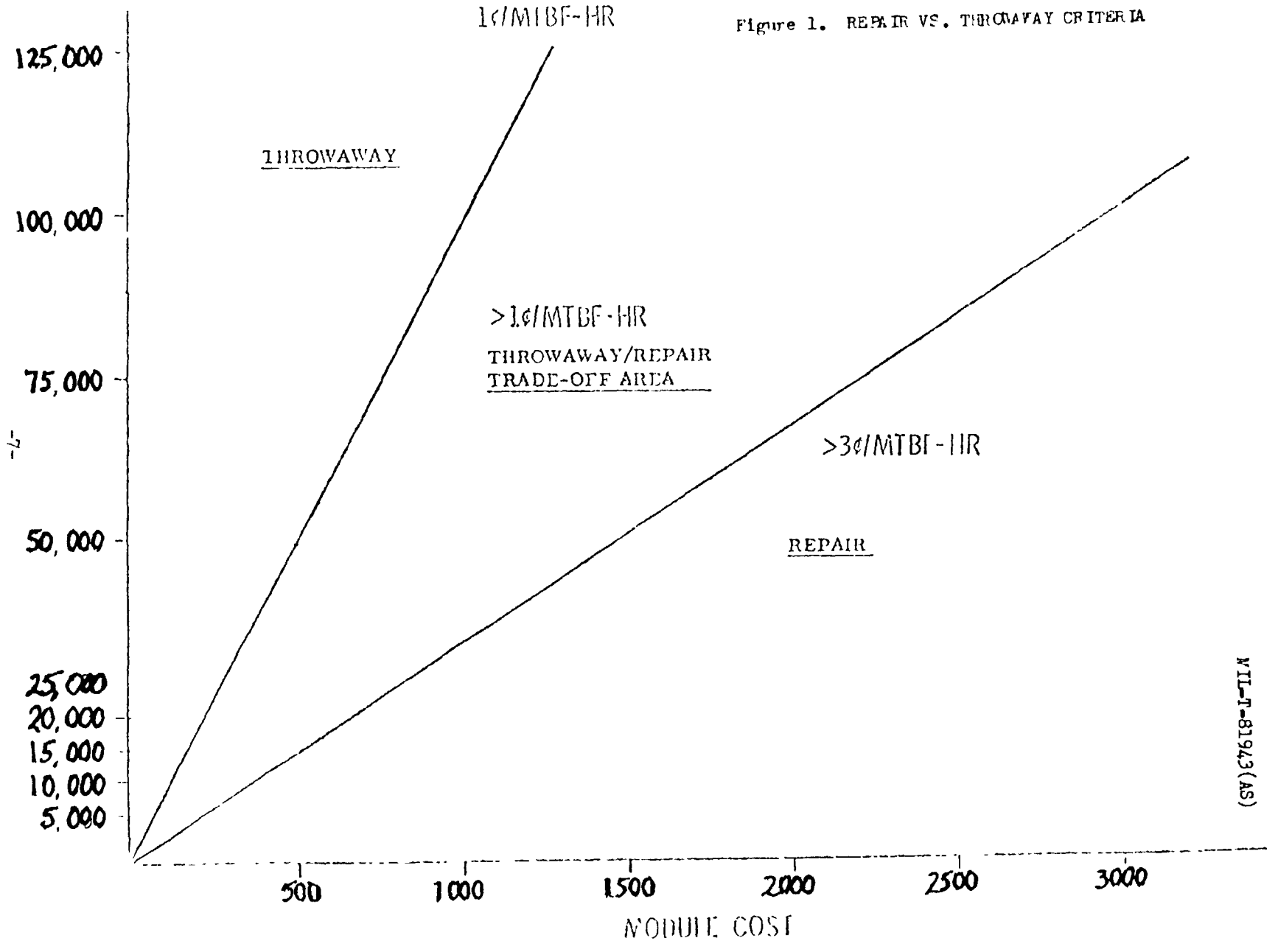


Figure 1. REPAIR VS. THROWAWAY CRITERIA

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3.3.7.7 Quick Replaceable Assembly - Replaceable assemblies shall, to the maximum extent possible, be designed as Quick Replaceable Assemblies as defined in AR-10, paragraph 6.1.5.

3.3.7.8 ERT (Equipment Repair Time) - The equipment shall meet the following:

3.3.7.8.1 Corrective Maintenance - Shall not exceed an ERT of one hour when tested in accordance with Method (3) of MIL-STD-471.

3.3.7.8.2 Preventative Maintenance - Exclusive of calibration cycles shall not exceed one hour for all preventative maintenance actions per maintenance cycle.

3.3.8 Nomenclature and Nameplates - Nomenclature assignment and nameplates approval for equipment identification shall be in accordance with the requirements of Specification MIL-N-18307.

3.3.9 Standard Conditions - The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests.

Temperature	Room Ambient (30° ±10°C)
Altitude	Normal Ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Voltage	115 ±1V, 400 Hz ±5 Hz, Single Phase

3.3.10 Service Conditions - The equipment shall operate satisfactorily under any of the environmental service conditions or reasonable combination of these conditions as specified in Specification MIL-T-21200 for Class 3 equipment, except as modified herein except that the lowest operating temperature shall be -20°C.

3.3.11 Warm Up Time - After a warm-up time of 5 minutes, under standard conditions or 15 minutes under service conditions, the performance specified shall be obtained and maintained for a least 300 hours continuously without adjustment of any controls not available on the front panel.

3.3.12 Primary Power Requirements - The equipment shall meet applicable requirements of MIL-T-21200 and shall give specified performance from the following power sources with characteristics as defined in MIL-T-21200 having limits specified herein.

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3.3.12.1 Single Phase Alternating Current Power - Within a frequency range of 50 to 420 Hertz with voltages between 103.5 and 126.5 volts rms.

3.3.12.2 Voltage Protection - The equipment shall not be damaged when exposed to transient voltages as defined in MIL-STD-704, and shall automatically resume normal operation when voltage returns within limits of normal operation.

3.3.13 Programmable Operation - The functions determined by the front panel controls as defined in paragraph 3.5.1.5.1.23 and 3.5.2.5.1.7 shall be digitally programmed via lines from the front panel controls or from receptacles on the rear of the unit provided to accept remote programming input signals and connected to the programming section receptacles. Four rear panel connectors shall be provided to accept programming inputs. Three rear panel connectors on the Power Supply - Video Pulse shall be MS3122E-22-55S-X,Y,Z connectors. The rear panel connector on the RF Signal Generator - Power Meter shall be MS3122E-20-41S-W connectors. These connectors shall be protected against adverse climate conditions with a cap which shall not protrude when in place, from the rear surface of the test set case. Programming inputs shall be discrete level signals. Discrete inputs shall be either logic "0" (0.0 to +0.45 volts DC) or logic "1" (+4.0 to +4.6 volts). When remote inputs are applied to the programming input connectors, operation of the test set shall be determined by signals from the programming source. With no inputs applied to the programming input connectors, operation of the test set shall be manual by means of the front panel control or as determined by front panel selectable internal programs. The programmable functions are listed below:

Range	135 Hz Modulation On/Off/Rev
Bearing	Identity Tone On/Off/Key
Mode	Squitter Rate
Pulse Spacing	A/A Pulse Repetition Rate
Reply Efficiency	Main Burst
Phase Shift	Auxiliary Burst
Range Tracking	Channel
Bearing Tracking	Output Level
15 Hz Percent Modulation	Pulse Modulation
135 Hz Percent Modulation	Antenna Speed
15 Hz Modulation On/Off/Rev	Pulse Rise Time
North Reference Trigger	Amplitude Modulation

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An additional internal board receptacle identical to those of the internal programming section shall be provided for mounting of logic for potential growth capability.

3.3.13.1 Standard Tests - The equipment shall be capable of being internally programmed to any one of four standard test conditions. These conditions shall be capable of being changed by card replacement. When the standard tests switch is in any internal program position and when the desired RF level has been selected via the level control switch, operation of the tests set shall be determined by signals from an internal programming source. With the standard tests switch in the manual position, operation of the test set shall be determined by means of the front panel controls. The standard conditions shall be internally programmable with the programs to be as follows:

Program Control Position

Function	1	2	3	4	5	6
Range		30.00	30.00	190	190	0
Bearing		90°	90°	90°	90°	0
Mode: Normal or Air-to-Air		Nor	Nor	A/A	A/AB	A/A
Mode: (X) or (Y)		X	X	X	X	X
Δ Pulse Space (from nominal)		0	0	0	0	0
Reply Eff		70	70	90	80	100
Phase Shift		0	0	0	0	0
Range Rate		0	0	0	0	0
Bearing Rate		0	0	0	0	0
% Mod 15 Hz		10	10	0	10	10
% Mod 135 Hz		10	0	0	0	10
15 Hz Mod		ON	ON	OFF	ON	ON
135 Hz Mod		ON	OFF	OFF	OFF	ON
ID Tone		KEY	OFF	OFF	OFF	OFF
Squitter Rate		2700	1000	0	1300	0
A/A PRF		OFF	OFF	1050	200	OFF
Main Burst		ON	OFF	OFF	ON	OFF
Aux Burst		ON	OFF	OFF	OFF	OFF
Channel		63	63	63	63	Manual
Output Level		See paragraph 3.5.1.5.1.23				
Pulse Modulation		Int	Int	Int	Int	Int
Antenna Speed		0	0	0	0	0
Pulse Rise Time		2.0	2.0	2.5	2.5	2.5
North Ref Trigger		OFF	ON	OFF	OFF	OFF

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Program Control Position	1	2	3	4	5	6

Program Modes

1. Manual
2. Rec - T/R
3. Inverse Receive
4. A/A Range Only
5. A/A Range and Bearing
6. Internal Self-Test

Level Control Position (-dbm)	0	10	77	80	87	90

3.4 Performance - The performance of the test set shall be as set forth below. Unless otherwise specified, these requirements shall apply under both standard and extreme service conditions.

3.4.1 Functions - The test set is used for the maintenance of airborne TACAN equipments and simulates a surface beacon in the X or Y mode of operation, a DME only beacon plus an airborne inverse mode receiving antenna, normally utilized by the set under test, or an airborne TACAN equipment operating in the following modes:

Normal A/A(X), Normal A/A(Y), Inverse A/A(X), Inverse A/A(Y)
Normal A/A BCN(X), Normal A/A BCN(Y), Inverse A/A BCN(X), Inverse A/A BCN(Y) modes.

It shall be capable of performing the following functions as required in the modes selected:

3.4.1.1 Measure the range accuracy of the airborne R/T unit and test its ability to search and track at different range rates and reply efficiencies.

3.4.1.2 Measure the bearing accuracy of the airborne R/T unit and test its ability to search and track at different bearing rates. This shall include "15 cycle only" operation (135 Hz modulation disabled). This shall also include testing of "Rec, T/R, A/A, and BCN inverse modes of operation" wherein an appropriately synchronized "main reference burst trigger pulse (video) output" shall be provided by the test set to enable the set to internally generate a "main reference burst" required for bearing measurement and as a component of the RF output signal

3.4.1.3 Test the operation of the airborne R/T unit identity circuit.

3.4.1.4 Measure the airborne receiver sensitivity, selectivity and adjacent channel rejection on all channels.

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3.4.1.5 Measure the peak power output of the airborne transmitter on all channels.

3.4.1.6 Measure and indicate reply efficiency of the test set and of the airborne set in the A/A (air-to-air) mode.

3.4.1.7 Measure and indicate percent modulation.

3.4.1.8 Measure and indicate the zero dbm power output level of the test set.

3.4.1.9 Accept interrogation pulses and/or modulation pulses and range delays externally provided and transmit corresponding reply pulses to permit testing of airborne TACAN sets utilized for stationkeeping and/or data link functions.

3.4.1.10 Measure and display air-to-air transponding delay time of the airborne unit under test.

3.4.1.11 Provide a properly synchronized logic switch for activation of the dead time and main reference burst features of the set under test as required for the 15 Hz bearing and inverse TACAN operation.

3.4.2 RF Output

3.4.2.1 Frequencies - Radio frequency output signals shall be provided to cover the frequency range of 962 through 1213 MHz in 1 MHz steps as follows: The output frequency for the normal X mode shall begin at channel 1 with 962 MHz and increase in 1 MHz increments until channel 63 is reached at 1024 MHz. Continuing, channel 64 will be at 1151 MHz and increase in 1 MHz increments until channel 126 is reached at 1213 MHz. The corresponding input frequency for the X mode shall begin with channel 1 at 1025 MHz and increase in 1 MHz increments until channel 126 is reached at 1150 MHz.

The Output Frequency for the Y Mode shall begin at channel 1 with 1088 MHz and increase in 1 MHz increments until channel 63 is reached at 1150 MHz. Channel 64 will be at 1025 MHz. The output frequency shall increase in 1 MHz increments therefrom until channel 126 is reached at 1087 MHz. The corresponding input frequency for the Y mode shall begin at 1025 MHz (related to channel 1 output frequency) and increase in 1 MHz increments until 1150 MHz is reached (related to channel 126 output frequency).

The Output Frequency for the A/A-X (Air-to-Air X) and A/A-Y (Air-to-Air Y) Modes shall begin at channel 1 with 1025 MHz and increase in 1 MHz increments until channel 126 is reached at 1150 MHz. The corresponding input frequency

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for the (A/A-X) and (A/A-Y) modes shall begin at 1088 MHz (related to channel 1 output frequency) and increase in 1 MHz increments until 1150 MHz is reached (related to channel 63 output frequency). Continuing, the corresponding input frequency shall be 1025 MHz (related to channel 64 output frequency) and increase in 1 MHz increments until 1087 MHz is reached (related to channel 126 output frequency).

3.4.2.2 Accuracy - Output frequencies shall be derived through frequency synthesis from a single crystal oscillator. Each output frequency shall be accurate to ±.001%. The accuracy of the internal crystal oscillator shall determine the output frequency accuracy.

3.4.2.3 RF Output - The RF output level shall be adjustable in 1 db steps from zero to 110 db below 1 milliwatt. RF leakage from cases and cables, on frequency and within the allowable pulse spectrum, shall be sufficiently low level as to in no way interfere with testing of airborne TACAN sets throughout the range of levels specified in the preceding sentence. Internal leveling shall be provided to maintain the output level at 0 dbm plus or minus 0.5 db with zero output attenuation. The variable output attenuator shall be accurate to plus or minus (0.1 db +0.01 db/db) throughout an operating range of 0 to 110 db below 1 milliwatt.

3.4.3 Modulation Characteristics - The RF output shall be capable of being modulated by the pulse component and/or composite bearing signals specified in paragraphs 3.4.3.2 and 3.4.3.3, respectively. Output pulse characteristics shall be as specified in 3.4.3.1. Continuous wave operation and external amplitude and pulse modulation capability shall also be provided. External pulse modulation shall result in a pulse as specified in 3.4.3.1 when applied to external pulse modulation (Gaussian) input. When external modulation is applied to external pulse modulation (square) input, an RF output pulse with a rise and fall time of 0.1 microsecond and a pulse width determined by the external modulation pulse, shall result. External amplitude modulation by a sine wave of up to 10 kHz shall be capable of modulating the RF output with a distortion no greater than 10% at a modulation level of 50%.

3.4.3.1 Pulse Characteristics

3.4.3.1.1 Pulse Shape - The pulse envelope as detected by a linear detector shall have a rounded shape falling within the following limits:

3.4.3.1.1.1 Pulse Top - The instantaneous amplitude of the pulse shall not, at any instant between the point of the leading edge at 95% maximum amplitude and the point of the trailing edge which is 95% of the maximum amplitude, fall below a line which is 95% of the maximum voltage amplitude of the pulse and is parallel to the base line.

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3.4.3.1.1.2 Pulse Rise Time - The time required for the leading edge of the pulse to rise from 10 to 90% of its maximum voltage amplitude shall be 2.5 ±0.1 or 2.0 ±0.25 microseconds as determined by a front panel control.

3.4.3.1.1.3 Pulse Fall Time - The time required for the trailing edge of the pulse to fall from 90 to 10% of its maximum voltage amplitude shall be 2.5 ±0.5 microseconds.

3.4.3.1.1.4 Pulse Duration - The pulse duration which is measured between the points on the leading and trailing edges of the pulse which are 50% of the maximum voltage amplitude of the pulse, shall be 3.5 ±0.3 microseconds.

3.4.3.1.2 Pulse Droop - With no composite bearing signal amplitude modulation applied, any pulse in a pair or in a reference group shall not deviate from the average peak amplitude by more than ±0.08%.

3.4.3.1.3 Pulse Coding and Spacing

3.4.3.1.3.1 Single pulses or 17 selectable pulse pair spacings in 0.5 microsecond steps over the following ranges shall be provided:

Mode	Range of Spacing
Normal, X	12 ±4 µs
Normal, Y	30 ±4 µs
A/A, X	12 ±4 µs
A/A, Y	24 ±4 µs
A/A, BCN, X	12 ±4 µs
A/A, BCN, Y	24 ±4 µs

3.4.3.1.3.2 Manual selection of pulse spacing between 6 and 39.5 microseconds in 0.5 microsecond steps shall be provided by activation of a control which overrides the spacing selected by the means provided in paragraph 3.4.3.1.3.1.

3.4.3.1.3.3 Spacing accuracy shall be ±0.10 microsecond as measured at the 50% maximum voltage amplitude points.

3.4.3.2 Pulse Modulation - Modulating pulses shall consist of range replies, bearing reference pulse groups, an identification signal, squitter, air-to-air interrogations, and externally supplied modulation as defined below:

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3.4.3.2.1 Range Reply Signal - A paired pulse range reply signal shall be transmitted when the test set is interrogated in the normal mode. When in the air-to-air mode, interrogation shall initiate a single pulse reply. The dead time shall not be less than 50 microseconds nor more than 60 microseconds, unless the dead time is defeated with the control provided. The simulator shall reply to interrogations, the interval between which is greater than the dead time plus T. If the dead time is defeated, the simulator shall decode and transmit replies to interrogations, the interval between which is T or greater, where T is as defined below:

$$T = (Ro \pm 0.1) + 12.359\ (1 \pm .0001)D$$

D = Simulated Range, indicated in nautical miles

T = Delay time in microseconds at the RF input/output connector of the test set and is the sum of the zero mile delay, Ro ±0.1, and the additional delay (12.359) (1 ±.0001) (D) incurred by the range simulated

Range replies simulating distance from 0 to 300 nautical miles in 0.1 mile increments shall have a time delay between an interrogation and the corresponding range reply in accordance with the above formula.

For Normal X and Y Modes: T shall be measured as the delay time between the 50% amplitude point of the respective leading edges of the first interrogation pulse of a coded pair and corresponding point on the first RF reply pulse of a coded pair.

In the X or Y Air-to-Air Modes: T shall be measured as the delay time between the 50% amplitude point of the respective leading edges of the first interrogation pulse of a coded pair and the corrsponding transmitted RF reply pulse.

Ro = Zero Mile Range Delay

Ro	Mode
50 ±0.1 microseconds	Normal, X
62 ±0.1 microseconds	A/A, X
74 or 56 ±0.1 microseconds Internally selectable	Y
74 ±0.1 microseconds	A/A, Y

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The test set shall be designed and calibrated to have range reply delay characteristics in accordance with the above formula. When the specified value of Ro is employed for the mode selected, the delay time (T) at the RF Input/Output connector shall be within limits established by the formula for any range simulated from 0 to at least 300 miles. The specified range accuracy shall be provided for peak input power levels from 400 to 5000 watts. Properly coded range reply pulses shall be generated for input power levels from 100 to 5000 watts.

3.4.3.2.2 Main Reference Pulse Group - The 15 Hertz reference bearing signal shall consist of a group of 12 pairs of pulses, with the spacing between pairs established 30 ±0.10 microseconds for the X mode and shall consist of a group of 13 single pulses spaced 30 ±0.10 microseconds for the Y mode and 10 single pulses spaced 30 ±0.1 microseconds for the A/A (X) mode. In each case, the repetition rate of the group shall be 15 groups per second ±0.2%.

3.4.3.2.3 Auxiliary Reference Pulse Group - The 135 Hertz reference bearing signal shall consist of 6 pairs of pulses, with the spacing between pairs established as 24 ±0.10 microseconds for the X mode and shall consist of a group of 13 single pulses spaced 15 ±0.10 microseconds for the Y mode. In each case, the repetition rate of the group shall be 135 Hz which shall be synchronized with the main reference group; however, the auxiliary group which would otherwise coincide in time with the main reference group shall not appear in this position in the final output signal. No auxiliary reference pulse groups shall be transmitted for A/A or inverse operation testing.

3.4.3.2.4 Identification Signal - The identification signal shall consist of a series of paired pulses having a spacing as specified in 3.4.3.1.3 transmitted at a repetition rate of 1350 pulse pairs per second and synchronized so that the first pulse of an identity pair shall occur 740 ±50 microseconds after the first pulse of the preceeding auxiliary reference group. To preserve a constant duty cycle and to minimize bearing error during identity signals, an equalizing pair of pulses shall be transmitted 100 ±10 microseconds after each identity pair. Provision shall be made to permit identification signal selection in one of three modes as follows: ON, OFF, or KEYED. During internal keyed operation, the squitter and range reply pulses shall be replaced by the coded identity group pulses at a recurrence rate of once every 37.5 seconds ±10%. The code shall be transmitted as dots and dashes (International Morse Code) of identity pulses; the spaces between the dots and dashes being occupied by range reply and squitter pulses. The dots shall be 0.125 second ±10%, and the dashes shall be 0.375 second ±10%. Spaces before and after the letter group shall be 0.625 second ±10%. Maximum message length shall not exceed 7.5625 seconds.

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3.4.3.2.5 Squitter Rate - When ground-to-air is simulated, the signal shall have randomly distributed pulses which shall be selectable by front panel control from 0 to 2700 ±81 pulse pairs per second in steps of 100 ±3.0 pulse pairs per second before the reference groups are added. When the A/A beacon mode of operation is simulated the squitter rate shall be adjustable from 0 to 2700 ±81 single pulses per second in steps of 100 ±3.0 with the front panel control provided. The random pulses shall include any range replies. The spacing between any two consecutively transmitted random pulse pairs that are not range replies shall not be less than 60 ±10 microseconds.

3.4.3.2.6 Air-to-Air Interrogations - In the air-to-air mode, the output shall include pulse pair interrogations with a rate adjustable from 0 to 1999 pulse pairs per second. In addition, range replies shall be single pulses as specified in paragraph 3.4.3.2.1.

3.4.3.2.7 Precedence - In the normal mode, squitter, identification, and range reply signals must not appear during a main or auxiliary reference pulse group.

3.4.3.2.8 External Pulse Delay and Modulation Input - External pulse delay and modulation capabilities required for testing of airborne TACAN sets when used for stationkeeping or TACAN Data Link shall be accommodated by the test set.

3.4.3.3 15 and 135 Hz Composite Bearing Signal - The composite bearing signal shall amplitude modulate the radio frequency output pulse signals in the normal X and Y, and A/A X and Y bearing modes. Fifteen Hz and 135 Hz sine wave components shall be related and added to form the composite signal as specified in Military Standard-291, except that tolerances shall be as specified in this specification. The output bearing signal shall not be in error by more than 0.25° (azimuth). Separate controls shall be provided for disabling or adjusting the amplitude of the 15 and 135 Hz sine wave components.

3.4.3.3.1 Harmonic Content - The harmonic content of the 15 Hz and 135 Hz components of the radio frequency output signals, other than the ninth harmonic of 15 Hz and the tenth harmonic of 135 Hz, shall not exceed 10% of each component.

3.4.3.3.2 15 Hz Bearing Information - Fifteen Hz simulated bearing shall be determined by the time delay from the main reference pulse group to the positive slope point of inflection of the 15 Hz component of the composite signal through use of the following formula.

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$$B_{(15)} = \frac{T \pm 9}{185.1852\ (1 \pm .001)}$$

where $B_{(15)}$ is the simulated bearing represented by the 15 Hz component in degrees and T is the delay time in microseconds. 185.1852 = microseconds per degree of the 15 Hz sine wave = microseconds per degree of bearing. For X mode operation, the delay shall be measured from the 50% amplitude point on the leading edge of the tenth pulse of the main reference pulse group. For Y mode operation, the delay shall be measured from a point which is midway between the 50% amplitude points on the leading edges of the sixth and seventh pulses of the main reference pulse group. For the A/A(X) mode of operation, the delay shall be measured from a point which is midway between the 50% amplitude points on the leading edges of the sixth and seventh pulses of the main reference group.

3.4.3.3.3 135 Hz Bearing Information - Vernier simulated bearing information within 40° bearing sectors shall be determined by the time delay from the 50% amplitude point on the leading edge of the twelfth pulse of the auxiliary reference pulse group to the positive slope point of inflection of the 135 Hz component of the composite signal. This delay time, representative of degrees of phase shift of the 135 Hz sine wave, is considered here in terms of the time equivalent of degrees of the 15 Hz sine wave ($B_{(15)}$) to obtain the bearing simulated by the composite bearing signal, directly.

3.4.3.3.4 15 and 135 Hz Composite Bearing Information - The bearing simulated by the 15 and 135 Hz composite bearing signal shall be given by:

$$B_{(15)\ (135)} = 40\ (N-1) + \frac{T \pm 9}{185.1852\ (1 \pm .001)}$$

where $B_{(15)\ (135)}$ is the simulated bearing in degrees determined by the 15 and 135 Hz composite bearing signal. N is an integer which represents the 40° sector in which $B_{(15),\ (135)}$ is located, as determined from the 15 Hz simulated bearing, $B_{(15)}$ and T are the delay time in microseconds within that sector to the bearing $B_{(15),\ (135)}$.

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3.4.3.4 RF Pulse Spectrum - The energy level at points plus or minus 0.8 MHz from the frequency of the channel utilized shall be at least 60 db below the energy level at the channel frequency. The energy level at points plus or minus 2.0 MHz from the channel frequency shall be at least 65 db below the energy level at the channel frequency. Pulse spectrum shall be measured at the test set RF output connector.

3.4.3.5 Spurious Outputs - The spurious content of the output signal shall be at least 60 db down except for the allowable pulse spectrum distributed around the output center frequency. The CW output between coded pairs of pulses shall be at least 70 db below the peak level of the pulse for output levels of 0 to -50 dbm. The CW between RF output coded pulse pairs shall not be greater than -120 dbm at any lower output level.

3.4.4 Blanking Signal Input - The test set shall accept externally supplied blanking pulse signals at the blanking signal input jack. Properly timed occurrences of these pulses shall cause blanking of internally generated squitter and range reply pulses at the time of input of related externally supplied modulation pulses. The externally supplied blanking signal shall be a pulse having 1.0 plus or minus 0.1 microsecond rise time as measured between the 10 and 90% amplitude points of the leading edge. Pulse duration shall be 2 through 7400 microseconds, as measured from the 50% amplitude point of the leading edge to the 50% amplitude point of the trailing edge. Fall time shall be 1.0 plus or minus 0.1 microsecond, as measured between the 90 and 10% points of the trailing edge of the pulse. The pulse top shall not fall below a line which is 95% of the maximum voltage amplitude of the pulse at any instant between 95% maximum amplitude point of the leading edge and 95% maximum amplitude point of the trailing edge of the pulse. The amplitude of the blanking signal shall be between 5.0 and 10.0 volts.

3.4.5 Trigger, Sync, and Reference Pulse Groups Outputs - Shall be provided at output jacks as listed under paragraph 3.5.1.5.3.

3.4.6 External Modulation Input - The test set shall accept at the external pulse modulation jack, externally generated modulation pulses having the same rise, duration, fall and top characteristics as the externally supplied blanking signal of paragraph 3.4.4 and transmit related reply pulse delayed 16 $\pm$0.1 microseconds as measured from the 50% amplitude point of the leading edge of the externally supplied modulation pulse to the 50% amplitude point of the leading edge of the transmitted RF pulse, provided that properly related corresponding preceding blanking pulses are externally provided at the blanking signal input jack. The amplitude of the externally supplied modulation pulse shall be between 5.0 and 7.0 volts. The test set shall sense and be driven at the 50 $\pm$2% amplitude point of the leading edge of the externally supplied modulation pulse.

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3.4.7 Peak Power Measurement Accuracy - The output power of the airborne TACAN transmitter shall be measured to an accuracy of plus or minus 0.6 db over a power range of 0.4 to 5.0 kilowatts including the effects of the RF line and load utilized by the test set. The specified accuracy shall be maintained over the frequency range of 1020 to 1155 MHz, inclusive. The indicated power shall be the average peak pulse power and shall be equal to the rms power of a sine wave having amplitude equal to the average of the peaks of the pulses measured.

3.4.8 Interruptive Self-Test - A means of signal switching shall be provided to allow the test set to interrogate itself for self-test purposes. During interruptive self-test operation, the input to the test set receiver shall be disconnected for the RF input/output connector and the test set output signal shall be applied to the RF input of the test set receiver in lieu thereof. Under this condition standard internal bearing and range delays in microseconds shall be displayed on digital indicators indicative of zero degree bearing and zero miles range and shall measure to ±0.1 microsecond the delay time from the 50% maximum amplitude point on the leading edge of an internally generated pulse subsequently routed via the standard bearing and range delays of the test set through the test set transmitter and receiver with the end of the interval of measurement measured at the 50% maximum amplitude point of the leading edge of the pulse at time of arrival at the circuit point where the generated pulse was originally delivered.

3.4.9 North Reference Trigger

3.4.9.1 Air-to-Air Mode - In the Air-to-Air Mode, the North Reference Trigger shall be made available at a front panel connector; which shall cause the TACAN under test (in the beacon mode) to produce a main reference burst of 10 single pulses with the zero degree reference point with respect to the positive going zero crossing point of the 15 Hz modulation waveform when operating with a rotating antenna midway between the 50% amplitude points of the leading edge of the sixth and seventh pulses.

The North Reference Trigger shall be capable of shunting to ground a voltage of 6 ±1.0 volts through an impedance of 1.0 K ohms or greater. The shunt shall be applied for a period of 70 ±10 microseconds at a repetition rate of 15 Hz ±0.2%.

3.4.9.2 Inverse Mode - When the rotation of an airborne receiving antenna is simulated while testing a TACAN in the inverse mode, the 15 Hz bearing shall be determined by the time delay from a point in time which is 167 microseconds after the opening of the north reference trigger switch

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to the negative slope point of inflection of the 15 Hz modulating signal. For zero degree bearing to be displayed on the TACAN under test in the inverse mode, the north reference trigger switch shall open at a point in time 167 microseconds prior to the negative going zero crossing point of the 15 Hz modulating waveform.

The time delay of 167 microseconds following the opening of the north reference trigger switch is the time from a point two microseconds prior to the 50% amplitude point of the first pulse of the Main Reference Pulse Group in the A/A Beacon Mode to the point midway between the 50% amplitude points of the leading edges of the sixth and seventh pulses.

The determination of the 15 Hz bearing to the negative slope point of inflection of the 15 Hz modulation signal shall be accomplished by maintaining a fixed timing for the North Reference Trigger and inverting the phase of the 15 Hz modulation signal with the front panel control provided.

The bearing as defined above shall be computed according to the following formula:

$$B_{(15)} = \frac{T \pm 9}{185.1852\ (1 \pm .001)}$$

where $B_{(15)}$ is the simulated bearing represented by the 15 Hz component in degrees and T is the delay time in microseconds. The number 185.1852 equals microseconds per degree of bearing.

3.5 Detail Requirements - The functions of the Test Set, Radio, AN/ARM-156 shall be provided in two (2) cases which shall meet the following requirements:

3.5.1 The Power Supply - Video Pulse, PP-6730/ARM-156 shall meet the following requirements:

3.5.1.1 Function - The Power Supply - Video Pulse of the test set shall consist of a decoder, range and bearing simulator circuitry, pulse synchronization circuitry, and a power supply. The decoder interrogation shall be routed to the range and bearing simulator circuitry, which shall provide selectable modulation of the RF output signals. Synchronizing signals, triggers and reference burst outputs shall be provided at jacks for external use.

3.5.1.2 Form Factor - The maximum external dimensions of the Power Supply - Video Pulse when housed in its case shall not exceed 15 inches high by 21 inches wide by 21 inches deep, and shall have an EMI tight cover which will fit within the confines of the case or a standard

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19-inch relay rack. The unit when removed from its case shall consist of a front panel 14 inches high by 19 inches wide, and be fitted with the above EMI cover behind the front panel. The front panel shall be fitted with a minimum of 14 captive screws for securing it to its case, 8 of which will be used to secure the unit to a relay rack. The Power Supply - Video Pulse when mounted within its case shall mate with, fasten to, and stack with the RF Signal Generator - Power Meter, SG-980/ARM-156 and with units of existing AN/ARM-31 and AN/ARM-155.

3.5.1.3 Weight - The weight of the Power Supply - Video Pulse shall meet the requirements of paragraph 3.3.1.

3.5.1.4 Content - The Power Supply - Video Pulse shall contain the following circuits.

3.5.1.4.1 Decoder - Circuitry shall be included which responds to nominal input pulse spacings as follows.

Mode	Pulse Spacing
Normal, X	12.0 ±0.5 microseconds
Normal, Y	36.0 ±0.5 microseconds
Air-Air, X	12.0 ±0.5 microseconds
Air-Air, Y	24.0 ±0.5 microseconds
Air-Air Beacon (BCN)X	12.0 ±0.5 microseconds
Air-Air Beacon (BCN)Y	24.0 ±0.5 microseconds

The decoder shall not respond to pulse pairs with spacings deviating from nominal by more than 4 microseconds. In addition, the time delay from the 50% amplitude point on the leading edge of the first input pulse to the 50% amplitude point on the leading edge of the second input pulse of a pair shall be measured and displayed to the nearest 0.1 microsecond.

3.5.1.4.2 Range - Calibrated fixed range positions shall be provided to give precise distance information from 0 to at least 300 nautical miles in 0.1 nautical mile increments as specified in paragraph 3.4.3.2.1 A range clock frequency of 16.1825 MHz or a multiple thereof, shall be generated within the test set to determine the 0.1 mile range increment accuracy. This frequency shall have an accuracy of ±.001%. The simulated range between 0 and at least 300 miles shall be displayed in nautical miles nmi to the nearest 0.1 mile for both static and dynamic operation. The range clock output shall be available at a connector on the rear of the unit.

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3.5.1.4.3 Range Rate - A calibrated control shall be provided for slewing the simulated range in or out at a rate from 0 to 6000 knots over a range of 0 to at least 300 miles. The accuracy shall be $\pm(10 + 0.1R)$ knots where R is the indicated range rate in knots. The simulated range rate shall be displayed to an accuracy of ± 1 knot.

3.5.1.4.4 Reply Efficiency - A means shall be provided for adjusting the reply efficiency of the test set from zero to maximum in steps of 10% of maximum. When reference bursts and squitter pulses are being generated, maximum reply efficiency is a function of the mode of operation. When reference bursts and squitter pulses are disabled, the maximum reply efficiency shall be 100% from 0 to at least 300 miles range. The air-to-air reply efficiency of the airborne set under test shall be measured and displayed. The actual reply efficiency shall be measured and displayed to the nearest 1%.

3.5.1.4.5 Bearing - Calibrated fixed bearing positions shall be provided to give precise bearing information from 0 to 359.9° in 0.1° increments as specified in paragraph 3.4.3.3. The simulated bearing between 0 and 359.9° shall be displayed in degrees to the nearest 0.1° for both static and dynamic operation.

3.5.1.4.6 Bearing Rate - A calibrated control shall be provided for slewing the simulated bearing in either direction at a rate from 0 to 30° per second in steps of 1° per second to an accuracy of $\pm 0.1°$ per second over the full range of 360°. The simulated bearing rate shall be measured and displayed to an accuracy of $\pm 0.1°$ per second.

3.5.1.4.7 Sine Wave Modulation - Circuitry shall be included for generating 15 Hz and 135 Hz sine waves and adding them to form the composite bearing signal. The frequency accuracy of the 15 Hz and 135 Hz sine waves shall be as specified in paragraph 3.5.1.4.11 and their distortion shall be as specified in paragraph 3.5.1.4.10. Calibrated controls shall be provided for independently adjusting the modulation percentage of the 15 Hz and 135 Hz sine wave signals from 0 to 40% in 1% steps which shall have an accuracy of $\pm 1\%$. Zero % modulation shall be as specified in paragraph 3.4.3.1.2. The total percent modulation of the composite or component sine wave signals shall be measured internally from the modulated RF and displayed to an accuracy of $\pm 1.0\%$. A sample of both the 15 Hz and 135 Hz sine waves shall be made available at front panel connectors.

3.5.1.4.8 15 Hz Phase Shift - A calibrated control shall be provided for shifting the phase of the 15 Hz signal by 25° in either direction in half degree increments from the coincident 0° position of 15 and 135 Hz signals. A shift of the phase of the 15 Hz signal shall not alter the

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position of the main reference pulse group. The simulated 15 Hz phase shift shall be measured and displayed to an accuracy of ±0.1 degree.

3.5.1.4.9 Signal Switching - Controls shall be included for turning on or off the 15 Hz and 135 Hz signals independently. These controls shall also provide for reversing the phase of each signal independently.

3.5.1.4.10 Distortion - The total harmonic distortion of either component of the composite bearing signal shall not exceed 2%.

3.5.1.4.11 Antenna Speed - A means shall be provided for selecting either fixed or variable simulated antenna speed. When a fixed antenna speed is selected, the 15 Hz and 135 Hz sine wave frequencies shall be accurate to .005%. When a variable antenna speed is selected, the simulated antenna speed shall be capable of being varied not less than ±2% from 900 rpm in steps of 0.01% by a front panel control. This shall represent a variation not less than ±2% of the 15 Hz and 135 Hz sine wave frequencies. The difference between the variable 15 Hz and fixed 15 Hz signal frequencies shall be measured and displayed to an accuracy of .005%, related to variations from the fixed 900 rpm antenna speed and fixed 135 Hz signal.

3.5.1.4.12 Identification Generator - This unit shall generate identification signal pulses as specified in 3.4.3.2.4.

3.5.1.4.13 Air-to-Air Pulse Generator - A digital control shall permit adjustment of the air-to-air interrogation rate from 0 to 1999 pulse pairs per second as specified in paragraph 3.4.3.2.6. The interrogating pulse pair repetition frequency, PPRF, in the air-to-air mode shall be measured and displayed to an accuracy of ±1 pulse pair per second.

3.5.1.4.14 Squitter Generator - A calibrated control shall permit the selection of squitter rates as specified in paragraph 3.4.3.2.5.

3.5.1.4.15 Encoder - This unit shall generate the pulse pair spacing specified in paragraph 3.4.3.1.3 and the reference pulse group spacing specified in paragraphs 3.4.3.2.2 and 3.4.3.2.3.

3.5.1.4.16 Mode Selector - A control shall be provided for selecting any one of the six modes specified in paragraph 3.4.1. The operating mode affects the output frequency as specified in paragraph 3.4.2.1, the output pulse pair spacing as specified in paragraph 3.4.3.1.3, the reference pulse group spacing as specified in paragraphs 3.4.3.2.2 and 3.4.3.2.3, and the decoder operation as specified in paragraph 3.5.1.4.1. In addition,

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operation in the A/A X range only mode or the A/A Y range only mode, inhibits the 15 Hz and 135 Hz signals specified in paragraph 3.4.3.3 and all pulse modulation signals specified in paragraph 3.4.3.2 except the range reply signal and the air-to-air interrogations. Operation in the A/AB(X) (Air-to-Air X Bearing and Range Mode) or the AABY (Air-to-Air Y Bearing and Range Mode) inhibits the 135 Hz signals specified in paragraph 3.4.3.3.3 and the pulse signals of paragraphs 3.4.3.2.3 and 3.4.3.2.4. Operation in the normal T/R Mode inhibits only A/A interrogating signals specified in paragraph 3.4.3.2.6. During simulated data link operation, utilizing an externally provided blanking signal, the presence of the blanking signal at the appropriate input jack of the test set shall inhibit coincidental internally generated range replies or squitter pulses.

3.5.1.4.17 A/A System Delay Time - The fixed delay of the unit under test shall be measured and displayed at least within the range of 57 to 79 microseconds from the 50% amplitude point of the leading edge of the first interrogation pulse to the 50% amplitude point of the leading edge of the reply pulse. The resolution and accuracy shall be $\pm$0.1 microsecond.

3.5.1.4.18 Test Signals - Sine wave signals of 15 Hz and 135 Hz, Main Reference Pulse Group, and Auxiliary Reference Pulse Groups shall be available at separate BNC connectors on both the front and rear panels.

3.5.1.4.19 Digital Display - Two five digital displays and selector switches shall be provided to indicate Range in increments of 0.1 mile, Bearing in increments of 0.1°, A/A Delay in increments of 0.1 microsecond, RF Input Pulse Spacing in increments of 0.1 microsecond, Reply Efficiency in increments of 1%, A/A Interrogation Frequency in increments of 1 pulse pair per second, Bearing Rate in increments of 1.0° per second and Range Rate in increments of 1 knot, change in Antenna Speed from nominal (15 RPS = 15 Hz) in increments of 0.0015 Hz and 15 Hz Phase Shift in increments of 0.5°.

3.5.1.4.20 Power Supply - A power supply to furnish the proper operating voltages shall be contained within the Power Supply - Video Pulse.

3.5.1.4.21 Power Interlock - Provisions shall be made for input power connections on both the front and rear panel. When power connections are made at the front panel connector, the rear panel connector shall be disconnected; and if the rear connector is used, the front panel power connector shall be discontinued. If both power connectors are connected the rear power connector shall take precedence.

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3.5.1.4.22 Data Link Pulses - A means shall be provided for generating data link RF pulse groups having single pulse characteristics as specified in paragraph 3.4.3.1, and data link video pulse groups as specified in paragraph 3.5.1.4.22. These data link pulse groups shall consist of three pulses spaced 18 ±0.1 microseconds apart. The spacing between the groups of three pulses shall be selectable from 60 to 490 microseconds in 10 microsecond steps to an accuracy of 1.0 microsecond when in either the Test Set Transmit or Airborne TACAN Transmit Modes. The spacing between the groups of three pulses shall be selectable from 100 to 490 microseconds in 10 microsecond steps to an accuracy of 1.0 microsecond when in the Bilateral Transmit Mode. The number of groups of pulses during a "Frame" shall be selectable from 0 to 399 in one pulse group steps to an accuracy of ±1.0 pulse group. The number of frames per second shall be selectable in number from 1 to 5.

Internal triggers for the generation of data link RF pulse groups and the data link video pulse groups selected shall be negative going pulses from a level of 4 ±1.5 vdc to a level of 0 ±.45 vdc and shall be made available at a front panel connector for external use. With respect to the internal trigger, the data link RF pulse groups so provided shall be delayed by 8 ±2.0 microseconds, and the data link video pulse groups shall be delayed by 50 ±0.5 microseconds.

A Data Link Enable Pulse shall be generated and provided at a front panel connector and shall bracket each related front panel provided data link video pulse group. The data link enable pulse shall have a level of 0 ±0.45 vdc during the bracketed interval and a level of 4 ±1.5 vdc during the non-bracketed interval.

A Data Link Precedence Pulse shall be generated and provided at the front panel Sync Connector. When selected by the Sync Control, the pulse shall bracket each related front panel provided group of triggers for the generation of data link RF pulse groups. The data link precedence pulse shall have a level of 0 ±0.45 vdc during the nonbracketed interval and a level of 4 ±1.5 vdc during the bracketed interval.

3.5.1.5 Front Panel - The front panel of the Power Supply - Video Pulse shall contain the following:

3.5.1.5.1 Controls - The following controls shall be located on the front panel of the Power Supply - Video Pulse.

3.5.1.5.1.1 Range - Digital controls shall be provided to permit the selection of simulated range from 0 to 399 miles in 0.1 mile increments.

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3.5.1.5.1.2 Bearing - Digital controls shall be provided to permit the selection of simulated bearing from 0 to 359.9° in 0.1° increments.

3.5.1.5.1.3 Mode - Digital controls shall be provided to select any one of six modes as follows: X or Y, G/A (Ground-to-Air), or A/A (Air-to-Air), AAB (Air-to-Air Bearing and Range) Modes as required to test airborne TACAN sets operation in the related modes listed in paragraph 3.4.1.

3.5.1.5.1.4 Pulse Spacing - Digital controls shall be provided to permit varying of the output spacing of pulses in a pair:

3.5.1.5.1.4.1 From nominal minus 4.0 microseconds to nominal plus 4.0 microseconds in 0.5 microsecond steps;

3.5.1.5.1.4.2 From 6 to 39.5 microseconds in 0.5 microsecond steps.

A switch position shall also be provided for the selection of single pulses.

3.5.1.5.1.5 Reply Efficiency - A digital control shall be provided to permit adjustment of the Reply Efficiency from zero to maximum in steps of 10% of maximum. Maximum reply efficiency shall be 100% in the absence of main or auxiliary reference, identity, A/A interrogations, data link, or externally generated data pulses.

3.5.1.5.1.6 Phase Shift - Digital controls shall be provided to permit adjustment of the phase of the 15 Hz component of the sine wave modulating signal from -25° to +25° in 0.5° steps.

3.5.1.5.1.7 Range Tracking - A calibrated digital control shall be provided to permit adjustment of simulated Range Rate from 0 to 6000 knots in 100 knot increments. A digital control shall provide selection of track in either the positive or negative directions, or no tracking of simulated range.

3.5.1.5.1.8 Bearing Tracking - A calibrated digital control shall be provided to permit adjustment of simulated Bearing Rate from 0 to 30° per second in 1° per second increments. A digital control shall provide selection of tracking in either the positive or negative directions, or no tracking of simulated bearing.

3.5.1.5.1.9 Percent Modulation - Two calibrated digital controls shall be provided to permit adjustment of the 15 Hz and 135 Hz sine wave modulation signals from 0 to 40% independently in 1% steps.

3.5.1.5.1.10 15 Hz Modulation - A digital control shall be provided for activating or removing the 15 Hz modulation signal or reversing phase.

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3.5.1.5.1.11 135 Hz Modulation - A digital control shall be provided for activating or removing the 135 Hz modulating signal or reversing phase.

3.5.1.5.1.12 Identity Tone - A digital control shall be provided to permit selection of ON, OFF, or KEYED coded identity tone operation.

3.5.1.5.1.13 Squitter Rate - A digital control shall be provided to permit the selection of squitter rate from 0 to 2700 pulse pairs per second in steps of 100 pulse pairs per second.

3.5.1.5.1.14 A/A PRF - Digital controls shall be provided to permit adjustment of the A/A Interrogating Pulse Repetition Frequency from 0 to 1999 pulse pairs per second in increments of one pulse pair per second. This control shall also permit inhibiting the A/A interrogation signal.

3.5.1.5.1.15 Main Burst - A digital control for removing the Main Reference Pulse Group shall be provided.

3.5.1.5.1.16 Auxiliary Burst - A digital control for removing the Auxiliary Reference Pulse Group for special test purposes shall be provided.

3.5.1.5.1.17 Antenna Speed - Digital controls shall be provided to permit adjustment of simulated antenna speed (15 Hz to 900 rpm) over a range of not less than ±2% of 15 Hz in steps of 0.01% (equals .0015 Hz).

3.5.1.5.1.18 Sync Select - A six position selector switch shall be provided to permit the selection of one of six signals to be connected to the sync output connector.

3.5.1.5.1.19 Display Select - Two selector switches shall be provided to permit the selection of the monitor signal to be applied to separate related digital displays.

3.5.1.5.1.20 Dead Time - A digital control shall be provided to permit defeat of dead time which normally follows an interrogation by the airborne set under test.

3.5.1.5.1.21 Power - A power ON/OFF control shall be provided.

3.5.1.5.1.22 Data Link - Digital controls shall be provided to permit selection of Data Link Pulse Group Spacing from 60 to 490 microseconds in 10 microsecond steps. Additionally, digital controls shall be provide to permit the selection of the number of Pulse Groups Per Frame from 0 to 399. A digital control shall be provided to permit the selection of one through five Frames Per Second. A three position toggle switch shall be

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provided to control the data link transmissions for either TS Trans (Test Set Transmit) or airborne TCN Trans (TACAN Trans) or BILAT (Bilateral Transmission) by controlling the application of the data link enable pulse to either the internal modulator, external connector, or both.

3.5.1.5.1.23 Internal Program/Self-Test - A digital control(s) shall be provided to permit the selection of either manual front panel control of this simulator, or one of four internally programmed standard test conditions, or interruptive self-test. A digital control(s) shall be provided to permit the selection of the following RF output levels when any standard internal program is selected.

(1) 0 dbm
(2) -10 dbm
(3) -77 dbm
(4) -80 dbm
(5) -87 dbm
(6) -90 dbm

All controls except (18) through (24) shall be capable of being programmed.

3.5.1.5.1.24 Internal/External Range Clock - A toggle switch shall be provided on the rear panel to permit the selection of either internal or external range clock signals.

3.5.1.5.1.25 North Reference Trigger - A digital control shall be provided for the ON/OFF operations of the north reference trigger.

3.5.1.5.2 Indicators - The following indicators shall be located on the front panel of the Power Supply - Video Pulse.

3.5.1.4.2.1 Digital Display - A minimum of two digital displays shall be provided to indicate the value of digital signals. The signals being indicated are determined by the positions of the related display select switches.

3.5.1.5.2.2 Power - A lamp shall be provided to indicate when power is applied to the test set.

3.5.1.5.2.3 Internal Alarm - A lamp shall be provided to indicate programmed operation of the test set.

3.5.1.5.2.4 Remote Alarm - A lamp shall be provided to indicate remote programmed operation of the test set.

3.5.1.5.2.5 Time Totalizing Meter - A meter shall be provided to indicate the cumulative on time of the Power Supply - Video Pulse.

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3.5.1.5.3 Electrical Connections - Connection to the Power Supply - Video Pulse from external circuits shall be provided as follows:

3.5.1.5.3.1 Power Input - A connector shall be provide to accept 115 ±10% VAC, 50 to 420 Hz, 1 phase primary input power.

3.5.1.5.3.2 Pulse - A BNC coaxial connector shall be provided to permit monitoring of the pulses applied to the modulator in the test set.

3.5.1.5.3.3 Sync - A BNC coaxial connector shall be provided that shall supply a sync pulse output signal to an oscilloscope. The sync pulse appearing at this connector shall be determined by the position of the sync selector switch.

3.5.1.5.3.4 Range Trigger - A BNC coaxial connector shall be provided that shall supply a trigger pulse for each reply pair transmitted by the test set.

3.5.1.5 3.5 Main Reference Pulse Group - A BNC connector shall be provided that shall supply Main Reference Pulse Groups that shall precede the related Main Reference Pulse Groups transmitted by the test set by not more than 10.0 microseconds.

3 5.1.5.3.6 Auxiliary Reference Pulse Group - A BNC coaxial connector shall be provided that shall supply Auxiliary Reference Pulse Groups that shall precede the related Auxiliary Reference Pulse Groups transmitted by the test set by not more than 10.0 microseconds.

3.5.1.5.3.7 Zero Degree Trigger - A BNC coaxial connector shall be provided that shall supply a trigger pulse for each positive slope point of inflection of the 15 Hz sine wave. The time of occurrence of this trigger pulse shall not be altered when the 15 Hz sine wave is phase shifted within the test set.

3.5.1.5.3.8 15 Hz Sine Wave - A BNC coaxial connector shall be provided to permit monitoring of the 15 Hz sine wave signal applied to the modulator in the test set.

3.5.1.5.3.9 135 Hz Sine Wave - A BNC coaxial connector shall be provided which permits monitoring of the 135 Hz sine wave signal applied to the modulator in the test set.

3.5.1.5.3.10 Data Link Enable - A BNC coaxial connector shall be provided that shall supply the data link enable pulse as described in paragraph 3.5.1.4.22 for external use.

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3.5.1.5.3.11 Data Link Pulse Groups - BNC coaxial connector shall be provided that shall supply the data link pulses generated as described in paragraph 3.4.1.4.22 for external use.

3.5.1.5.3.12 North Reference Trigger Switch - A BNC coaxial connector shall be provided that shall supply the North Reference Trigger as described in paragraph 3.4.9 for external use.

3.5.1.5.3.13 Range Clock Input/Output - A BNC coaxial connector shall be provided which permits monitoring of the Range Clock Output Frequency or permits input from an external range clock.

3.5.1.5.3.14 Programming Inputs - Three multipin connectors shall be provided which shall accept the inputs required for programmable operation.

3.5.1.5.3.15 Display Inputs - A multipin connector shall be provided to accept digital inputs from external equipment for display on the 5-digit digital displays. The display inputs shall be discrete level signals. Discrete inputs shall be either logic "0" (0.0 to 0.45 volts DC) or logic "1" (+4.0 to 4.6 volts DC). Logic "0" level is defined as the "true" state and logic "1" level is defined as the "false" state.

3.5.1.5.3.16 Locations of Connectors - The connectors specified in paragraphs 3.5.1.5.3.13 through 3.5.1.5.3.15 shall be located on the rear panel. The connectors specified in paragraphs 3.5.1.5.3.1 through 3.5.1.5.3.12 shall be located on the front panel with a duplicate set of these connectors located on the rear panel.

3.5.2 RF Signal Generator - Power Meter - The RF Signal Generator Power Meter SG980/ARM-156 shall meet the following requirements:

3.5.2.1 Function - The RF Signal Generator - Power Meter of the test set shall consist of a receiver, RF signal generator, modulator and a peak power monitor. The receiver shall accept the interrogations from the airborne set under test and provide these signals to the Power Supply - Video Pulse and at output jacks. The RF Signal Generator and Modulator shall supply modulated RF signals to the airborne TACAN set under test. Provisions shall be made for application of externally generated modulation to the signal generator. The power meter shall measure and indicate the fundamental peak power output of the airborne TACAN transmitter under test.

3.5.2.2 Form Factor - The maximum external dimensions of the RF Signal Generator - Power Meter, when housed in its case shall not exceed 9.25 inches high by 21 inches wide by 21 inches deep, and shall have an EMI tight cover which shall fit within the confines of the case or a standard 19-inch relay rack. The unit when removed from its case shall consist of a front panel 8.75 inches high by 19 inches wide, and

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fitted with above EMI cover behind the front panel. The front panel shall be fitted with a minimum of 10 captive screws for securing it to its case, 4 of which will be used to secure the unit to a relay rack. The RF Signal Generator - Power Meter, when mounted within its case shall mate with, fasten to, and stack Power Supply - Video Pulse, PP-6730/ARM-156 and units of existing AN/ARM-31 and AN/ARM-155.

3.5.2.3 Weight - The weight of the RF Signal Generator - Power Meter shall meet the requirements of paragraph 3.3.1.

3.5.2.4 Content - The RF Signal Generator - Power Meter shall contain the following circuits.

3.5.2.4.1 RF Signal Generator - The RF signal generator shall consist of the following:

3.5.2.4.1.1 Frequency Synthesizer - The output frequencies specified in 3.4.2.1 shall be obtained from a frequency standard through frequency synthesis. The Frequency Selector control shall be calibrated to read directly in channel numbers. The channel number indicated on the test set dial shall be that of the equipment wnose operation is simulated and the related output frequency provided shall be in agreement therewith for all modes of operation. Output frequency shall be determined by both the channel selector and mode selector positions.

3.5.2.4.1.2 RF Circuits

3.5.2.4.1.2.1 Modulator - The modulators utilized shall provide the modulation characteristics specified in paragraph 3.4.3. The pulse modulating signal shall be obtained from the circuitry specified in paragraph 3.5.2.4.1.5. The composite sine wave modulating signal shall be obtained from the circuitry specified in paragraph 3.5.1.4.7.

3.5.2.4.1.2.2 Output Level - The output level into the airborne TACAN receiver shall be variable from 0 to 110 db below 1 milliwatt (0 dbm to -110 dbm) in 1 db steps. The Output Level Control shall be calibrated to read directly in db below 1 milliwatt (dbm), (0 dbm = 1 milliwatt).

3.5.2.4.1.2.3 Diplexer - Interrogations to and replies from the test set appear at a single RF input/output connector. Routing of interrogating signals from the airborne TACAN transmitter to the test set receiver and dummy load and routing of the test set replies to the airborne TACAN receiver shall be provided by a diplexer.

3.5.2.4.1.2.4 Receiver - The receiver shall convert interrogating signals from the airborne TACAN transmitter to video pulses. Dynamic range and delay characteristics shall be such that the requirements of 3.4.3.2.1 shall be met when the test set is connected to the airborne TACAN set under test with the RF connecting means provided.

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3.5.2.4.1.3 Test Set Monitor - Circuitry shall be provided to sample and demodulate the test set RF output signal. The resultant video signal shall appear at the Test Set Monitor connector mounted on the front panel.

3.5.2.4.1.4 Interrogation Monitor - Circuitry shall be provided to sample and demodulate the RF output of the airborne TACAN transmitter. The resultant video signal shall appear at the Interrogation Monitor connector mounted on the front panel. The delay time from the RF input connector to the interrogation monitor connector shall be 0.2 ±0.1 microseconds.

3.5.2.4.1.5 Pulse Modulation - Circuitry shall be included for shaping the reference burst and pulse pair triggers to provide the modulated pulse shape specified in paragraph 3.4.3.1. A sample of this pulse shall be made available at a front panel connector. A selector switch shall be provided to permit Internal or External Pulse Modulation or CW operation.

3.5.2.4.1.6 Digital Display - A four digit display and control shall be provided to indicate peak power in watts to an accuracy of ±0.5 db with minimum display resolution of ±1% of the measured value, percent modulation in increments of 1% and power supply voltages to an accuracy of 1%.

3.5.2.4.2 RF Power Measurement Unit

3.5.2.4.2.1 Function - The RF Power Measurement Unit shall be removable from the front panel of the test set and shall measure and indicate the fundamental peak power output of the airborne TACAN transmitter under test to an accuracy of ±0.5 db. The level of measured power shall be displayed on the digital display of the RF Signal Generator - Power Meter. This unit shall be capable of being rigidly mounted to the airborne TACAN set under test to eliminate flexible cable insertion loss variations. Required mounting adapters shall be provided for each type of TACAN set to provide direct solid coaxial connection and mechanical support.

3.5.2.4.2.2 Form Factor - The RF Power Measurement Unit shall be enclosed in a case for use when the unit is removed from the test set and shall have maximum dimensions not to exceed 6 inches high by 5.5 inches wide by 7 inches deep.

3.5.2.4.2.3 Weight - The weight of the RF power measurement unit shall not exceed 10 pounds.

3.5.2.4.2.4 Content - In addition to the power sampling device, the RF power measurement unit shall contain:

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3.5.2.4.2.4.1 Low Pass Filter - A low pass filter shall be provided to attenuate the harmonics of the fundamental frequency of the signal to be measured at least 30 decibels before measurement by the RF Power Measurement Unit. The insertion loss of the filter shall not exceed plus or minus 0.2 db from 1020 through 1160 MHz. VSWR of the filter shall not exceed 1.2 to 1.0 from 1020 through 1160 MHz.

3.5.2.4.2.4.2 RF Load - A precision RF load of 50 ohms impedance, having a VSWR of 1.1 to 1.0 or less from 1020 to 1160 MHz, which shall satisfactorily dissipate 5.0 kilowatts peak and 15.0 watts average power, shall be provided.

3.5.2.5 Front Panel - The front panel of the RF Signal Generator - Power Meter shall contain the following:

3.5.2.5.1 Controls - The following controls shall be located on the front panel of the RF Signal Generator - Power Meter.

3.5.2.5.1.1 Channel - Digital control(s) shall be provided to permit the selection of any one of 126 TACAN channels, representative of 252 channel frequencies dependent upon the concurrent positions of the mode selector switches.

3.5.2.5.1.2 Output Level - Digital controls shall be provided to permit adjustment of the test set output level from 0 to -110 dbm in 1.0 db steps (0 dbm = 1 milliwatt).

3.5.2.5.1.3 Pulse Modulation - A five position switch shall be provided which permits selection of CW, Internal, External, Internal or External, or External Square Pulse Modulation. Signal precedence in the "Internal or External" position shall be controlled by an external blanking signal.

3.5.2.5.1.4 Display Select - A digital control shall be provided to permit the selection of the monitor signal applied to the digital display.

3.5.2.5.1.5 Pulse Rise Time - A two position toggle switch shall be provided to select either 2.5 or 2.0 microsecond pulse rise time.

3.5.2.5.1.6 Amplitude Modulation Select - A two position toggle switch shall be provided on the rear panel to select internal or external amplitude modulation.

3.5.2.5.1.7 All controls except paragraph 3.5.2.5.1.7 shall be capable of being programmed.

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3.5.2.5.2 Indicators - The following indicators shall be located on the front panel of the RF Signal Generator - Power Meter.

3.5.2.5.2.1 Digital Display - A digital display shall be provided to indicate the value of digital signals. The signal being indicated is determined by the position of the Display Select Control.

3.5.2.5.2.2 Time Totalizing Meter - A meter shall be provided to indicate the cumulative ON TIME of the RF Signal Generator - Power Meter.

3.5.2.5.3 Electrical Connections - Connections to the RF Signal Generator - Power Meter from external circuits shall be provided as follows:

3.5.2.5.3.1 RF Input/Output - An HN coaxial connector shall be provided to receive interrogations from and transmit replies to the unit under test. See paragraph 3.5.2.4.2.1.

3.5.2.5.3.2 Video Monitor

3.5.2.5.3.2.1 Interrogation Monitor - A BNC coaxial connector shall be provided to permit monitoring the interrogations and replies from the airborne set under test. Delay time from the RF input/output connector to this connector shall be 0.2 ±0.1 microseconds.

3.5.2.5.3.2.2 Test Set Monitor - A BNC coaxial connector shall be provided to permit monitoring of the interrogations and replies from the test set.

3.5.2.5.3.3 External Gaussian Pulse Modulation - A BNC coaxial connector shall be provided to permit gaussian pulse modulation of the test set from an external source. Delay time from the external pulse modulation input connector to the output RF input/output connector shall be 16 ±0.1 microseconds.

3.5.2.5.3.4 Internal Clock Frequency - A BNC coaxial connector shall be provided to permit monitoring of the frequency synthesizer reference oscillator of 37.5 MHz.

3.5.2.5.3.5 External Square Pulse Modulation - A BNC coaxial connector shall be provided to permit square pulse modulation of the test set from an external source. Delay time from the external pulse modulation input connector to the output RF input/output connector shall be 0.110 ±0.05 microsecond.

3.5.2.5.3.6 External Amplitude Modulation - A BNC coaxial connector shall be provided to permit amplitude modulation of the test set from an external source.

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3.5.2.5.3.7 External Blanking Input - A BNC coaxial connector shall be provided via which related internal circuits shall accept an externally provided signal for blanking of internally generated squitter and range replies, per paragraph 3.4.6.

3.5.2.5.3.8 RF Power Measurement Unit Connector - Multipin and RF connectors shall be provided to supply power to the RF power measurement unit and connect its output to the digital display selector and RF signal generator circuits.

3.5.2.5.3.9 Location of Connectors - The connectors specified in paragraphs 3.5.2.5.3.2.1 and 3.5.2.5.3.2.2, 3.5.2.5.3.8 and 3.5.2.5.3.11 shall be located on the front panel. The connectors specified in paragraphs 3.5.2.5.3.3 through 3.5.2.5.3.7 shall be located on the rear panel.

3.5.2.5.3.10 Programmable Inputs - A multipin connector shall be provided on the rear of the unit which shall accept the inputs for programmable operation.

3.5.2.5.3.11 Internal Gaussian Modulation Pulse - A BNC coaxial connector shall be provided to permit monitoring of the internal gaussian modulation pulse.

3.5.3 Transit Case - The requirements of MIL-T-21200 shall apply.

3.5.3.1 Transit Case, Power Supply - Video Pulse - The maximum dimensions of the Power Supply - Video Pulse Transit Case shall be 28 inches in length by 28 inches in width by 26 inches in height. The maximum weight of the transit case shall be 55 pounds.

3.5.3.2 Transit Case, RF Signal Generator - Power Meter - The maximum dimensions of the RF Signal Generator - Power Meter Transit case shall be 26 inches in length by 24 inches in width by 18 inches in height. The maximum weight of the transit case shall be 55 pounds.

3.5.4 Cable Assemblies and Accessories

3.5.4.1 Cables and Adapters - The following cables and adapters shall be supplied for use with the Test Set Radio, AN/ARM-156:

Ref Designation	Function
W-1	Input Power
W-2	Power Interface
W-3	Signal Interface
W-4	RF Input/Output
MS-2193	400 Hz Adapter for use in conjunction with cable W-1

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3.5.4.2 Accessories - A Printed Circuit Module Extender shall be furnished with and stored internally within the test set to aid in the maintenance of the test set.

3.6 Workmanship - Workmanship shall be in accordance with MIL-T-21200.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the government. The government reserves the right to perform any of the inspection set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Classification of Tests - Items covered by this specification shall be subjected to the following tests to determine compliance with all applicable requirements.

(1) Preproduction Tests
(2) Initial Production Tests
(3) Acceptance Tests
(4) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on an equipment representative of the production equipments to be supplied under the contract. Preproduction tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of paragraph 4.6. The government inspector and the procuring activity shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commerical testing laboratory acceptable to the government.

4.2.1 Preproduction Test Data - The contractor shall submit all data collected in conducting these tests to the procuring agency for review and approval. On the first production contracts this data shall include a list of all electrical and electronic parts.

4.2.2 Scope of Tests - Preproduction tests shall include all tests deemed necessary by the procuring activity to determine that the equipment meets all the requirements of Specification MIL-T-5422 to the limits specified in Specification MIL-T-21200, except as modified herein (see paragraph 3.3.10).

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4.2.3 Preproduction Approval - Approval of the preproduction sample shall be by the procuring activity upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction sample. Prefabrication of production equipment prior to the approval of the preproduction sample is at the contractor's own risk. The approved preproduction sample shall be retained by the contractor for his use in the fabrication and testing of equipment submitted for acceptance. The preproduction sample shall not be considered as one of the equipments under the contract.

4.2.4 Production Equipments - Equipments supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be equivalent to the approved preproduction sample. Each equipment shall be capable of successfully passing the same tests as imposed on the preproduction samples. Evidence of noncompliance with the above shall constitute cause of rejection and for equipment already accepted by the government, it shall be the obligation of the contractor to make necessary corrections as approved by the procuring activity.

4.3 Initial Production Tests - One of the first ten production equipments shall be selected and sent at the contractor's expense to a designated government laboratory for tests. This equipment shall be selected by the procuring activity after the equipment has successfully passed all individual tests (see paragraph 4.4.1). The preproduction sample shall not be selected for this test.

4.3.1 Scope of Tests - This equipment may be subjected to any and all tests the procuring activity deems necessary to assure that the production equipment is equivalent to the previously approved preproduction sample in design, construction, workmanship, performance and quality and that it meets all applicable requirements.

4.3.2 Accessory Material - In addition to the complete equipment submitted for initial production tests the contractor shall also submit such accessory material and data necessary to test the equipment.

4.3.3 Initial Production Sample Approval - Approval of the initial production sample shall be by the procuring activity upon satisfactory completion of all tests. Any design, material or performance defect made evident during this test shall be corrected by the contractor to the satisfaction of the procuring activity. Failure of the initial production sample to pass any of the tests shall be cause for deliveries of equipment under the contract to cease until proper corrective action is approved and accomplished. Corrective action shall also be accomplished on equipment previously accepted when requested by the procuring activity.

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4.3.4 Reconditioning of Initial Production Test Sample - On completion of the initial production test the equipment shall be reworked by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.4 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the government inspector. Contractors not having testing facilities satisfactory to the procuring activity shall engage the service of a commercial testing laboratory acceptable to the procuring activity. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of materials during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following.

(1) Individual Tests,
(2) Sampling Tests,
(3) Reliability Assurance Tests, and
(4) Special Tests.

4.4.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product,
(2) Operational Test, and
(3) Manufacturing Run-in Test.

4.4.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the materials and workmanship requirements have been met.

4.4.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.4.1.3 Manufacturing Run-in Test - Each equipment shall be operated under the conditions specified herein for a period of 6 hours without failure. A failure shall be defined as anything which causes mal-

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functioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments that are accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration	Any selected frequency within the range of 20 to 30 Hz (excluding resonant points) and a minimum amplitude of 3g.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6 hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to the plane for 5 minutes. During the 6 hour period of operation following the 10 minute vibration period, the equipment shall be mechanically cycled periodically through its various phases of operation. Should a failure occur, it shall be repaired and the test started over, except that the 10 minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failure occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures.

4.4.2 Sampling Tests - Equipment selected for sampling tests shall first have passed the individual tests. Equipments shall be selected for sampling tests by the government inspector in accordance with the following: (Sampling tests shall not be conducted unless reliability assurance tests of paragraph 4.4.3 are deleted by contractual action.)

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Sampling Tests
First 10	1
Next 50	2
Next 75	2
Next 100	2
1 for each additional 200 or fraction thereof	

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NOTE: If by contract action the initial production test (paragraph 4.3) is deleted, then a sample test shall be conducted on two equipments from the first 10 produced.

4.4.2.1 Scope of Tests - As a minimum each equipment selected by sampling tests shall be subjected to the following tests:

(1) Complete operational tests at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Operational test at certain environment conditions. The conditions may vary for each equipment tested and should be based on results of the preproduction, initial production, individual and special tests.

(3) Manufacturing run-in test specified in paragraph 4.4.1.3 except that the test duration shall be 120 hours. Each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurrence in future equipment.

Should repetitive failures occur, corrective action shall be required to eliminate the defect. Special tests (paragraph 4.4.4) shall be initiated to satisfy the procuring activity that the defect has been corrected.

4.4.3 Reliability Assurance Tests - Reliability assurance tests shall be conducted using MIL-STD-781. Tests shall be conducted as requred by the qualification phase and the production acceptance sampling phase.

4.4.3.1 Production Qualification Phase - Prior to the acceptance of equipments under the contract or order, a minimum of three equipments shall be tested as outlined in MIL-STD-781, under the section entitled "Qualification Phase of Production Reliability Tests." The maximum number of equipments to be used shall be those listed in table V of MIL-STD-781. For the qualification phase test plan IV, test level E shall be used except that -54°C shall be changed to -20°C; the "on" time of any single temperature cycle shall be 20 hours; the "off" time of any single temperature cycle shall be 4 hours; and the rate of change of the thermal medium shall average not less than 2°C a minute.

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4.4.3.2 Reliability Production Acceptance Sampling Phase Test - Samples of the equipment shall be tested as outlined in MIL-STD-781, under the section entitled "Sampling Phase of Production Reliability Tests." For the sampling phase, test plan IV, test level E shall be used except that -54°C shall be changed to -20°C; the "on" time of any single temperature cycle shall be 20 hours; the "off" time of any single temperature cycle shall be 4 hours; and the rate of change of the thermal medium shall average not less than 2°C a minute.

4.4.3.2.1 Lot Size for Sampling Phase - The entire contract quantity (with the exception of those equipments used in the qualification phase) shall be considered one lot unless otherwise specified in the contract. The first four available production units from the lot shall constitute the sample to be tested unless otherwise specified in the contract. The samples shall be tested until an accept or reject decision is reached. In the meantime other equipment which have passed individual tests of paragraph 4.4.1 shall be shipped. The test results shall be summarized monthly for the procuring activity. The procuring activity reserves the right to stop the acceptance of equipment any time after a reject decision has been reached, pending a review of the contractor's efforts to improve the equipment, the equipment parts and the equipment quality control.

4.4.4 Special Tests - Special tests shall be conducted on a quantity of equipment for the purpose of checking the effect of any design or materials change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling or reliability assurance tests.

4.4.4.1 Special Test Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early equipment after an engineering or material change.

(2) Whenever failure reports or other information indicate additional tests are required. (This will be determined by the procuring activity.)

4.4.4.2 Scope of Tests - Special tests shall consist of such tests as approved by the procuring activity. Test procedures previously approved to the preproduction tests shall be used where applicable. When not applicable the contractor shall prepare a test procedure and submit it to the procuring activity for approval prior to conducting the tests.

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4.4.5 Equipment Failure - Should a failure occur during either the sampling, reliability assurance or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the procuring activity for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the indivdual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.5 Life Test - The contractor shall furnish all samples and shall be responsible for accomplishing the life test. The test shall be of 300 hours duration and shall be conducted on equipments that have passed the individual tests. The life test sample shall be performed under the conditions specified in paragraph 4.5.1. The life test sample shall be selected by the government inspector in accordance with the following. (Equipments which have successfully passed the initial production tests, sampling tests, reliability tests, or special tests may be selected for life tests.) When reliability tests are conducted, the life test may be omitted if, during the reliability tests, a quantity of equipments equal to, or more than, that listed below receive at least 300 hours each of test time.

Quantity of Equipments Offered for Acceptance	Quantity to be Selected For Life Test
First 25	1
Next 50	1
Next 100	1 for each addition 300 or fraction thereof

4.5.1 Test Conditions - The life test shall be conducted under the following simulated service conditions.

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Temperature	Room Ambient (30° ±10°C)
Altitude	Normal Ground (0 - 5000 ft)
Humidity	Room Ambient
Input Voltage	115 ±10% (at lowest applicable frequency)
Vibration	None

4.5.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.5.3 Performance Check - At approximately 8 hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the procuring activity.

4.5.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all tube failures. This record shall indicate the following:

(1) part type number
(2) the circuit reference symbol number
(3) the part function
(4) name of manufacturer
(5) nature of the failure
(6) the number of hours which the part operated prior to failure

4.5.4.1 Failure Report - In the event of a failure, the government inspector shall be notified immediately. A report shall be submitted to the procuring activity upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The procuring activity will review such proposals and determine whether they are acceptable.

4.5.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life test shall be reconditioned as follows:

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(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all "wear" items. The "wear" items shall be determined by agreement between the contractor and the procuring activity.

(2) After reworking the contractor shall resubmit the equipment for acceptance.

4.6 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the procuring activity for review and approval. The right is reserved by the procuring activity or the government inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification and the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved test procedures are available from previous contracts such procedures will be provided and may be used when their use is approved by the procuring activity. However, the right is reserved by the procuring activity to require modification of such procedures including additional tests, when deemed necessary.

4.7 Reconditioning of Tested Equipment - Equipment which has been subjected to acceptance and life tests shall be reconditioned by the contractor by replacing all worn or damaged items. After reworking, the contractor shall resubmit the equipment for acceptance.

4.8 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.9 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished to the government inspector.

5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with specification MIL-E-17555.

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6. NOTES

6.1 Intended Use - The beacon simulator specified by this document is intended for use in the maintenance, development, and production of airborne TACAN receiver-transmitters and related systems.

6.2 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of the specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-T-21200, it is desired that their use be investigated. When investigation shows advantages can be realized, a request for approval shall be submitted to the procuring activity for consideration. Each request shall be accomplished by complete support information.

6.3 As a general rule, nonrepairable Subassemblies should be encapsulated or hermetically sealed. The number of connections internal to the subassembly should be held to a minimum. Detail parts, tolerances and ratings should be so selected that the life of the subassembly is greater than that of a similar repairable one. With few exceptions (such as high voltage power supplied), the nonrepairable subassembly should evidence a Mean Time To Failure greater than 5000 hours, and for many applications this figure must be nearer 50,000 hours.

6.4 Precedence of Documents - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be specifically approved in writing by the procuring activity.

(3) Referenced Specifications - Any referenced specifications shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

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6.5 The parentheses (Y), when used in the type designation, will be deleted or replaced by either a number or letter furnished by the procuring activity upon application by the contractor for assignment of nomenclature in accordance with paragraph 3.3.8. The complete type number shall be used on nameplates, shipping records and instruction books, as applicable.

6.6 Ordering Data - Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(1) Title, number and date of this specification.

(2) Selection of applicable levels of packaging and packing (see paragraph 5.1).

(3) Data requirements (see paragraph 3.6).

6.7 Asterisk - In specification revisions and superseding amendments an asterisk "*" preceding a paragraph number denotes paragraphs in which changes have been made from the previous issue. This has been done as a convenience only and the government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content as written, irrespective of the asterisk notations and relationship to the last previous issue.

6.8 Associated Equipment - The equipment shall operate with the following associated equipment:

(1) Test Set Indicator, AN/ARM-31
OF
Test Set Indicator, AN/ARM-155

(2) VAST System or equivalent external program source.

6.9 Supercession Data - This specification supersedes:

(1) MIL-T-81580(AS) of 25 April 1968,
(2) MIL-T-19203C of 15 May 1965, and
(3) NAVAIRSYSCOM Spec AS-2699 of 1 February 1971

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DOCUMENT IDENTIFIER AND TITLE
Mil T-81943(AS) Test Set, Radio AN/ARM-156

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTEPPRETATION IN PROCUREMENT USE?
A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?
☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO

DATE

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